

ST. XAVIER'S COLLEGE

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LAB/WRITTEN ASSIGNMENT NUMBER: 4.2

“MICROSOFT EXCEL”

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Introduction to Spreadsheet Software

Labeled Blueprint of MS Excel Interface

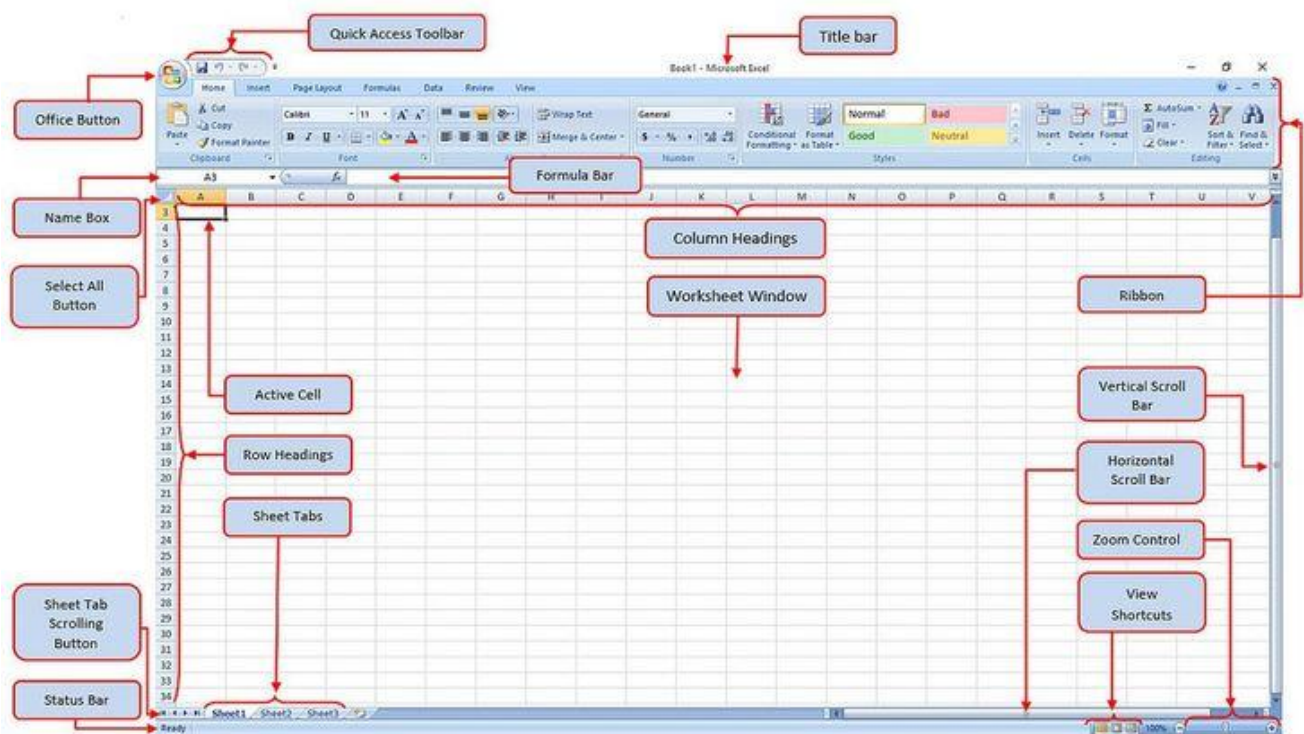


Figure 1: Structural Blueprint of the Microsoft Excel User Interface

Ribbon Tabs	Primary Purpose	Key functional Groups and functions	
Formula Tab	The primary purpose of the Formulas Tab is to perform calculations, automate mathematical tasks, and audit logic errors within your spreadsheet.	Function Library	Houses Excel's 400+ built-in equations organized by category for quick insertion.
		Defined Names	Allows you to create custom labels for cell ranges to make formulas easier to read.
		Formula Auditing	Troubleshoots errors by visually tracking the mathematical relationships between cells.
		Calculation	Controls when and how Excel updates the math in your sheet (automatically or manually).
Data Tab	The primary purpose of the Data Tab is to import, clean, organize, and analyze large datasets efficiently.	Get & Transform Data	Imports, merges, and cleans data from external sources like websites, databases, or CSVs.
		Sort & Filter	Reorganizes rows instantly based on criteria like alphabetical order, color, or specific values.
		Data Tools	Cleans up existing sheet data, separates combined text, and enforces data entry rules.
		Forecast / What-If Analysis	Simulates different financial or numerical scenarios to predict future outcomes.

Table 1: Detailed classification of Microsoft Excel Ribbon tabs, their corresponding subgroups, and core functions.

Some Definitions and Important Topics

Relative Cell Reference

A Relative Cell Reference is the default way Excel remembers cell locations. Instead of locking onto a specific grid coordinate, Excel remembers the distance and direction (e.g., "two cells to the left") from the cell containing the formula.

For example: A4, B1, C6,etc

Mixed Cell Reference

A mixed cell reference in a spreadsheet locks either the column or the row while keeping the other relative (e.g., \$A1 locks column A, while A\$1 locks row 1). When copied across cells, the coordinate with the dollar sign (\$) stays fixed, while the un-locked coordinate adjusts automatically.

Absolute Cell Reference

An absolute cell reference is a fixed cell address in a spreadsheet (written with two dollar signs, like \$A\$1) that remains completely unchanged when copied or dragged to other cells. The dollar signs lock both the column and the row coordinates.

VLOOKUP

VLOOKUP (Vertical Lookup) is a spreadsheet function used to search for a specific value in the first column of a data range and return a corresponding value from another column in the same row.

Syntax: =VLOOKUP(lookup_value, table_array, col_index_num, [range_lookup])

1. **Lookup_value:** The specific value, text, or cell reference you want to search for in your lookup table.
2. **Table_array:** The range of cells containing both your lookup data and the result data. Crucial Rule: The column containing the lookup_value must be the very first (leftmost) column in this range.

3. Col_index_num: The position number of the column you want to retrieve data from, counted from left to right starting at 1 for the first column of table_array.
4. Range_lookup: 0 or FALSE finds an Exact Match. Returns an error (#N/A) if the item isn't found and 1 or TRUE finds an Approximate Match. Used for numerical ranges, grades, or tax brackets (requires data to be sorted in ascending order).

Example =VLOOKUP(E2, A1:C4, 3, FALSE)

Task 1: Student Mark sheet Calculation

	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T
1	ST XAVIER'S COLLEGE																			
2	ROLL NO	NAME	SEC	PHY TH	PHY PR	CHEM TH	CHEM PR	MATH TH	MATH PR	ENG TH	ENG PR	NEP TH	NEP PR	COM TH	COM PR	TOTAL	REMARK	PERCENTAGE	GRADE	RANK
3	1	Peter Parker	A	56	23	55	22	69	23	55	21	67	22	48	45	506	PASS	84.33333333	A	3
4	2	Robert Downy JR	B	45	22	68	23	66	22	65	22	65	23	45	46	512	PASS	85.33333333	A	2
5	3	Chirs Hemsworth	I	48	24	44	22	54	23	53	23	58	21	46	47	463	PASS	77.16666667	B+	7
6	4	Scarlet Johnson	H	35	21	65	21	44	21	33	21	44	23	33	45	406	PASS	67.66666667	B	10
7	5	Vison	F	64	14	71	22	43	20	47	23	60	22	33	45	464	PASS	77.33333333	B+	6
8	6	Sang Chi	D	47	19	73	25	33	22	48	24	63	23	42	47	466	PASS	77.66666667	B+	5
9	7	Groot	E	23	24	45	24	43	25	55	25	65	24	41	48	442	FAIL	73.66666667	NG	NO RANK
10	8	Walter White	B	22	25	34	20	73	24	67	20	66	25	40	43	459	FAIL	76.5	NG	NO RANK
11	9	Jesse Pinkman	I	34	23	66	21	44	22	61	23	67	21	45	44	471	PASS	78.5	B+	4
12	10	Gustavo Fring	G	56	22	65	22	71	21	60	21	68	22	43	43	514	PASS	85.66666667	A	1

Figure 2 Student Marksheet Calculation

=Q5/600*100

Figure 3 Formula of Percentage

=SUM(E5:P5)

Figure 4 Formula of Total

=IF(R5="FAIL","NG",IF(S5>89,"A+",IF(S5>79,"A",IF(S5>69,"B+",IF(S5>59,"B",IF(S5>49,"C+",IF(S5>39,"C",IF(S5>29,"D+"))))))))

Figure 5 Formula of Grade

=IF(OR(E5<27,G5<27,I5<27,K5<27,M5<27,O5<15,F5<8,H5<8,J5<8,L5<8,N5<8,P5<15),"FAIL","PASS")

Figure 6 Formula of Remark

=IF(R5="FAIL","NO RANK",RANK(S5,\$S\$5:\$S\$14))

Figure 7 Formula of Rank

Task 2: VLOOKUP based result calculation

FIRST TERMINAL REPORT CARD 2026					
St. Xaviers College Maitighar, Kathmandu					
Roll No		1			
Student's Name		Peter Parker			
Section		A			

Subject	Theory Marks	Practical Marks	Obtained Marks in	Obtained Marks in	Remarks
1 Physics	75	25	56	23	PASS
2 Chemistry	75	25	55	22	
3 Mathematics	75	25	69	23	
4 English	75	25	55	21	
5 Nepali	75	25	67	22	
6 Computer	50	50	48	45	

Percentage Obtained:	84.33333333
Overall Grade	A
Rank	3

Grading system	
Percentage	Grade
90-100	A+
80-89	A
70-79	B+
60-69	B
50-59	C+
40-49	C
30-39	D+


 Exam Coordinator


 Vice Principal


 Principal

Figure 8 VLOOKUP based Mark sheet calculation

FIRST TERMINAL REPORT CARD 2026

St. Xaviers College
Maitighar, Kathmandu

Roll No	1
Student's Name	Peter Parker
Section	A

Subject	Theory Marks	Practical Marks	Obtained Marks in	Obtained Marks in	Remarks
1 Physics	75	25	56	23	PASS
2 Chemistry	75	25	55	22	
3 Mathematics	75	25	69	23	
4 English	75	25	55	21	
5 Nepali	75	25	67	22	
6 Computer	50	50	48	45	

Percentage Obtained:	84.33333333
Overall Grade	A
Rank	3

Grading system

Percentage	Grade
90-100	A+
80-89	A
70-79	B+
60-69	B
50-59	C+
40-49	C
30-39	D+


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St. Xaviers College
Maitighar, Kathmandu

Roll No	2
Student's Name	Robert Downy JR
Section	B

Subject	Theory Marks	Practical Marks	Obtained Marks in	Obtained Marks in	Remarks
1 Physics	75	25	45	22	PASS
2 Chemistry	75	25	68	23	
3 Mathematics	75	25	66	22	
4 English	75	25	65	22	
5 Nepali	75	25	65	23	
6 Computer	50	50	45	46	

Percentage Obtained:	85.33333333
Overall Grade	A
Rank	2

Grading system

Percentage	Grade
90-100	A+
80-89	A
70-79	B+
60-69	B
50-59	C+
40-49	C
30-39	D+


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Roll No	3
Student's Name	Chirs Hemsworth
Section	I

Subject	Theory Marks	Practical Marks	Obtained Marks in	Obtained Marks in	Remarks
1 Physics	75	25	48	24	PASS
2 Chemistry	75	25	44	22	
3 Mathematics	75	25	54	23	
4 English	75	25	53	23	
5 Nepali	75	25	58	21	
6 Computer	50	50	46	47	

Percentage Obtained:	77.16666667
Overall Grade	B+
Rank	7

Grading system

Percentage	Grade
90-100	A+
80-89	A
70-79	B+
60-69	B
50-59	C+
40-49	C
30-39	D+


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Roll No	4
Student's Name	Scarlett Johnson
Section	H

Subject	Theory Marks	Practical Marks	Obtained Marks in	Obtained Marks in	Remarks
1 Physics	75	25	35	21	PASS
2 Chemistry	75	25	65	21	
3 Mathematics	75	25	44	21	
4 English	75	25	33	21	
5 Nepali	75	25	44	23	
6 Computer	50	50	33	45	

Percentage Obtained:	67.66666667
Overall Grade	B
Rank	10

Grading system

Percentage	Grade
90-100	A+
80-89	A
70-79	B+
60-69	B
50-59	C+
40-49	C
30-39	D+


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Roll No	5	▼
Student's Name	Vishal	
Section	F	

Subject	Theory Marks	Practical Marks	Obtained Marks in	Obtained Marks in	Remarks
1 Physics	75	25	64	14	PASS
2 Chemistry	75	25	71	22	
3 Mathematics	75	25	43	20	
4 English	75	25	47	23	
5 Nepali	75	25	60	22	
6 Computer	50	50	33	45	

Percentage Obtained:	77.33333333
Overall Grade	B+
Rank	6

Grading system

Percentage	Grade
90-100	A+
80-89	A
70-79	B+
60-69	B
50-59	C+
40-49	C
30-39	D+

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Figure 9 VLOOKUP based mark sheet calculation of roll 1 to 5

Lab Activity Reflections

- I learned to automate data retrieval with VLOOKUP where I created the mark sheet based on the roll no of student.
- Learned to calculate total, rank , percentage and to handle data in MS Excel.
- Learned to use library functions effectively to calculate grade and to say if a student is pass or fail.

Conclusion

This lab provided hands-on experience in turning raw student academic records into a functional, automated reporting system. Mastering functions like VLOOKUP and GPA conditional logic demonstrated how spreadsheets save time, maintain accuracy, and transform complex numerical data into clear, professional mark sheets.